

CSE 150A-250A AI: Probabilistic Methods

Lecture 4

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof. Berg-Kirkpatrick)

Agenda

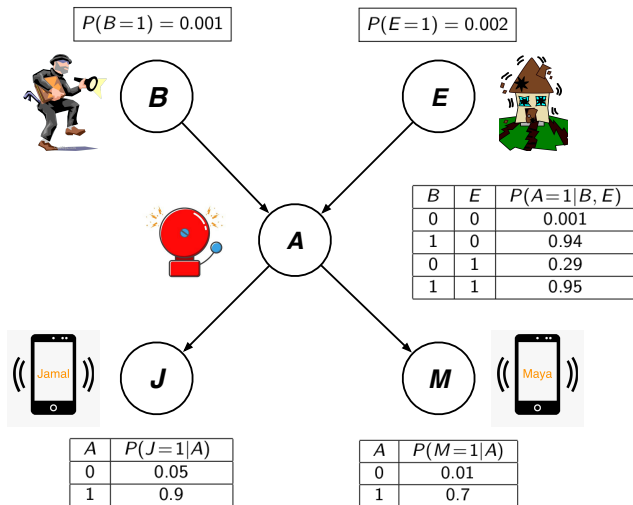
Review

Conditional probability tables

d-separation and examples

Review

Alarm example



A **belief network** (BN) is a directed acyclic graph (DAG) in which:

1. Nodes represent random variables.
2. Edges represent (direct) dependencies.
3. Conditional probability tables (CPTs) describe how each node depends on its parents.

$$\text{BN} = \text{DAG} + \text{CPTs}$$

Marginal and conditional independence in DAGs

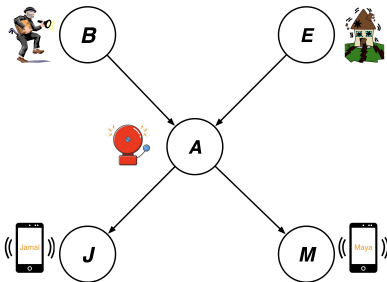
- Missing edges encode assumptions of independence:

$$P(X_i | X_1, \dots, X_{i-1}) = P(X_i | \text{pa}(X_i))$$

where $\text{pa}(X_i) \subseteq \{X_1, \dots, X_{i-1}\}$ denotes the **parents** of node X_i .

In words: Each variable is conditionally independent of its **non-descendants** given its **parents**.

- Alarm example:



$$P(E) = P(E|B)$$

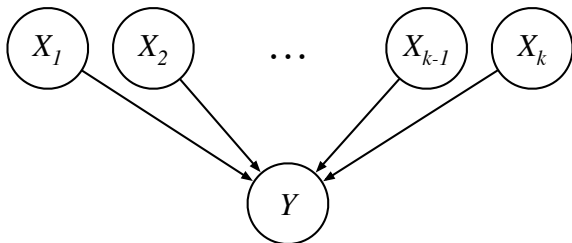
$$P(J|A) = P(J|A, B, E)$$

$$P(M|A) = P(M|A, B, E, J)$$

These are true no matter what CPTs are attached to the nodes in the DAG.

Conditional probability tables

Representing CPTs

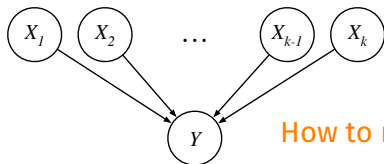


- How to represent $P(Y|X_1, X_2, \dots, X_k)$?
- Simplest case:

Suppose $X_i \in \{0, 1\}$, $Y \in \{0, 1\}$ are **binary** random variables.

How to represent $P(Y=1|X_1, X_2, \dots, X_k)$?

Types of CPTs

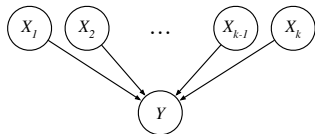


How to represent $P(Y=1|X_1, X_2, \dots, X_k)$?

Some possibilities:

1. Tabular
2. Logical / Deterministic
3. Noisy-OR
4. Sigmoid

1. Tabular CPT



X_1	X_2	$\dots$	X_k	$P(Y=1 X_1, X_2, \dots, X_k)$
0	0	$\dots$	0	0.1
1	0	$\dots$	0	0.6
0	1	$\dots$	0	0.3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	1	$\dots$	1	0.2

A lookup table can exhaustively enumerate a conditional probability for every configuration of parents.

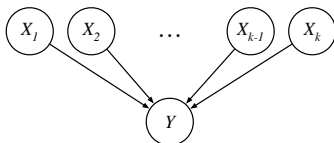
Pro

Able to model arbitrarily complicated dependence.

Con

A table with 2^k rows is too unwieldy for large k .

2. Logical / Deterministic CPT



CPTs can also mimic the behavior of logical circuits.

AND gate

$$P(Y=1|X_1, X_2, \dots, X_k) = \prod_{i=1}^k X_i$$

OR gate

$$P(Y=0|X_1, X_2, \dots, X_k) = \prod_{i=1}^k (1 - X_i)$$

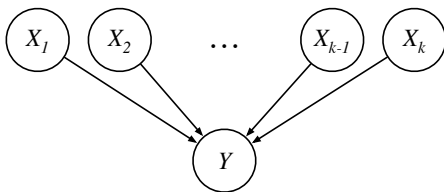
Pro

Compact representation for large k .

Con

No model of uncertainty.

3. Noisy-OR CPT



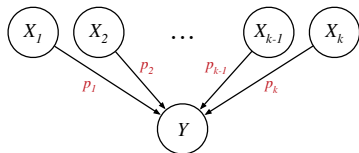
Use k numbers $p_i \in [0, 1]$ to parameterize all 2^k entries in the CPT:

$$P(Y=0|X_1, X_2, \dots, X_k) = \prod_{i=1}^k (1 - p_i)^{X_i}$$

$$P(Y=1|X_1, X_2, \dots, X_k) = 1 - \prod_{i=1}^k (1 - p_i)^{X_i}$$

But why is this called Noisy-OR?

Noisy-OR CPT (con't)



$$P(Y=1|X_1, X_2, \dots, X_k) = 1 - \prod_{i=1}^k (1 - p_i)^{X_i}$$

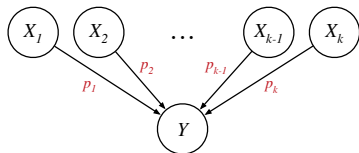
- When all parents are equal to zero:

$$P(Y=1|X_1=0, X_2=0, \dots, X_k=0) = 1 - \prod_{i=1}^k (1 - p_i)^0 = 1 - \prod_{i=1}^k (1) = 0$$

- When exactly one parent X_j is equal to one:

$$\begin{aligned} P(Y=1|X_1=0, \dots, X_{j-1}=0, X_j=1, X_{j+1}=0, \dots, X_k=0) \\ &= 1 - (1 - p_1)^0 \cdots (1 - p_{j-1})^0 (1 - p_j)^1 (1 - p_{j+1})^0 \cdots (1 - p_k)^0 \\ &= 1 - (1 - p_j) \\ &= p_j \end{aligned}$$

Noisy-OR CPT (con't)



$$P(Y=1|X_1, X_2, \dots, X_k) = 1 - \prod_{i=1}^k (1 - p_i)^{X_i}$$

- **Modeling uncertainty**

Intuitively, $p_i \in [0, 1]$ is the probability that $X_i=1$ **by itself** triggers $Y=1$.

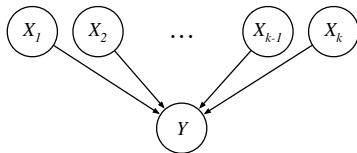
- **Logical OR as special case**

We recover a logical OR gate by taking the limit $p_i \rightarrow 1$ for all parents $i = 1, 2, \dots, k$.

- **Canonical application**

The parents $\{X_i\}_{i=1}^k$ are diseases, and the child Y is a symptom. The more diseases, the more likely is the symptom.

4. Sigmoid CPT

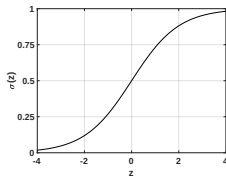


Use k real numbers $\theta_i \in \Re$ to parameterize all 2^k entries in the CPT:

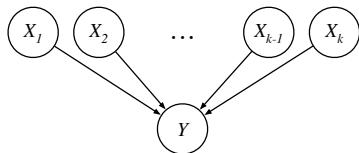
$$P(Y=1|X_1, X_2, \dots, X_k) = \sigma \left(\sum_{i=1}^k \theta_i X_i \right)$$

The function on the right hand side is called the **sigmoid** function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



4. Sigmoid CPT (con't)



$$P(Y=1|X_1, X_2, \dots, X_k) = \sigma \left(\sum_{i=1}^k \theta_i X_i \right)$$

Other uses of sigmoid functions:

- Activation function in neural nets
- Inverse of the link function for logistic regression

Properties:

- If $\theta_i > 0$, then $X_i=1$ favors $Y=1$.
- If $\theta_i < 0$, then $X_i=1$ inhibits $Y=1$.
- These effects can mix in a sigmoid CPT (unlike noisy-OR).

d-separation and examples

Conditional independence

- What we've already seen

A node X_i is conditionally independent of its non-parent ancestors given its parents:

$$P(X_i | X_1, X_2, \dots, X_{i-1}) = P(X_i | \text{pa}(X_i))$$

- What we can ask more generally

Let X , Y , and E refer to disjoint *sets* of nodes in a BN.

When is X conditionally independent of Y given E ?

$$\text{When is } \left\{ \begin{array}{l} P(X|E, Y) = P(X|E) \\ P(Y|E, X) = P(Y|E) \\ P(X, Y|E) = P(X|E) P(Y|E) \end{array} \right\} ?$$

- Above is special case

$$X = \{X_i\}, \quad E = \text{pa}(X_i) \quad Y = \{X_1, X_2, \dots, X_{i-1}\} - \text{pa}(X_i)$$

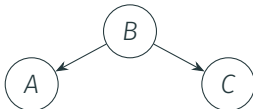
Base Cases

Causal chain



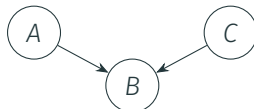
$$A \rightarrow B \rightarrow C$$

Common cause



$$A \leftarrow B \rightarrow C$$

Common effect



$$A \rightarrow B \leftarrow C$$

collider / v-structure

What happens when we observe the middle node B ?

Base Cases: Chain and Common Cause

Causal chain

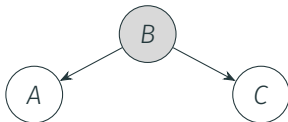


$$A \rightarrow B \rightarrow C$$

Observing B **blocks** the path.

$$A \perp C \mid B$$

Common cause



$$A \leftarrow B \rightarrow C$$

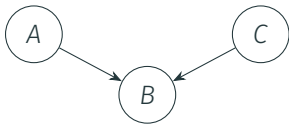
Observing B **blocks** the path.

$$A \perp C \mid B$$

What about a common effect?

Base Case: Common Effect

B unobserved

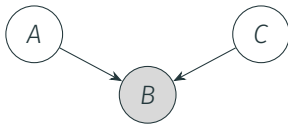


$$A \rightarrow B \leftarrow C$$

The collider **blocks** the path.

$$A \perp C$$

B observed



$$A \rightarrow B \leftarrow C$$

Observing the collider **opens** the path.

$$A \not\perp C \mid B$$

Colliders behave differently!

d-separation = direction-dependent separation

- Motivation

How is conditional independence in a BN encoded by the structure of its DAG?

- Theorem

$P(X, Y|E) = P(X|E)P(Y|E)$ if and only if every *path* from a node in X to a node in Y is *blocked* by E .

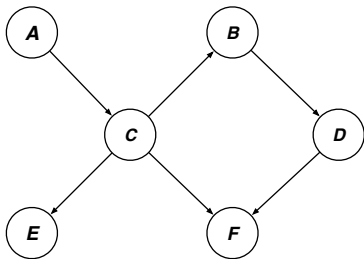
What counts as a path, and when is it blocked?

Paths in DAGs

- Definition

A **path** is any sequence of nodes connected by edges (*regardless of their directionalities*); it is also assumed that no nodes repeat.

- Examples

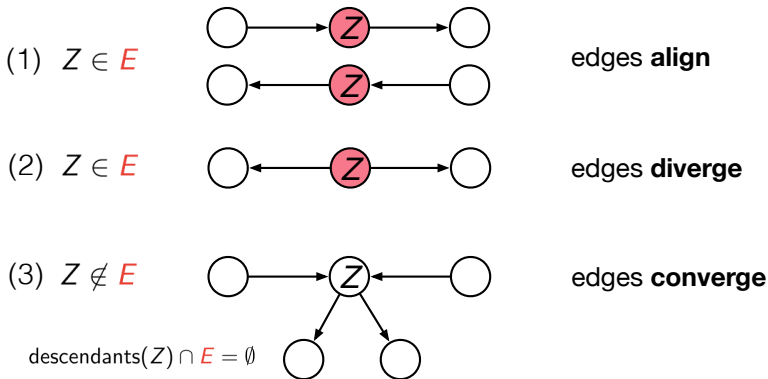


? paths from A to D:

Blocked paths

- Definition

A path π is **blocked** by a set of nodes E if there exists a node $Z \in \pi$ for which one of the three following conditions hold.

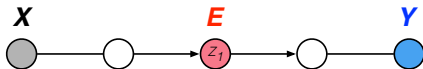


d-separation

- Theorem

$P(X, Y|E) = P(X|E)P(Y|E)$ if and only if every path from a node in X to a node in Y is *blocked* by E .

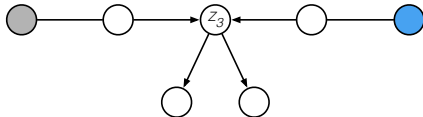
- Intuition



$Z_1 \in E$ is an intervening event in a causal chain



$Z_2 \in E$ is a common explanation or cause



$Z_3 \notin E$, $\text{desc}(Z_3) \cap E = \emptyset$ is an unobserved common effect

- Theorem

$P(X, Y|E) = P(X|E)P(Y|E)$ if and only if every *path* from a node in X to a node in Y is *blocked* by E .

- **Proof** (not given)

The proof of the theorem is non-trivial.

You are **not** responsible for its proof.

- **How useful is the theorem?** **Very!**

There are efficient algorithms to test d-separation in large BNs.
You should become skilled at these tests in simple BNs.

Alarm example

A. TRUE or B. FALSE?

1. $P(B|A, M) \stackrel{?}{=} P(B|A)$

The evidence is $\{A\}$.

There is one path $B \rightarrow A \rightarrow M$.

Node A satisfies condition (1).

The statement is .

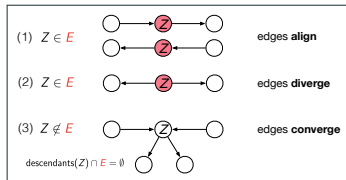
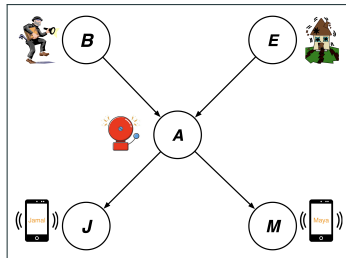
2. $P(J, M|A) \stackrel{?}{=} P(J|A) P(M|A)$

The evidence is $\{A\}$.

There is one path $J \leftarrow A \rightarrow M$.

Node A satisfies condition (2).

The statement is .



Alarm example (con't)

A. TRUE or B. FALSE?

3. $P(B) \stackrel{?}{=} P(B|E)$

The evidence is $\{ \}$.

There is one path $B \rightarrow A \leftarrow E$.

Node A satisfies condition (3).

The statement is .

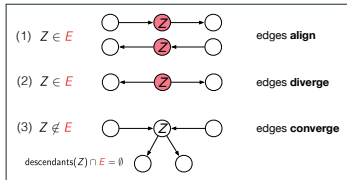
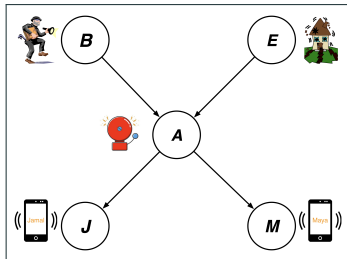
4. $P(B|M) \stackrel{?}{=} P(B|M, E)$

The evidence is $\{M\}$.

There is one path $B \rightarrow A \leftarrow E$.

Note that $M \in \text{desc}(A)$.

The statement is .



Loopy example

A. TRUE or B. FALSE?

5. $P(B|D, E) \stackrel{?}{=} P(B|D)$

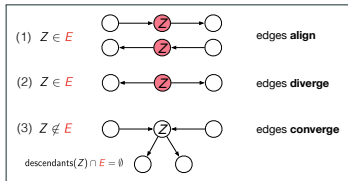
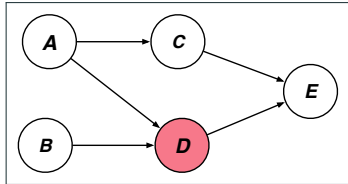
The evidence is $\{D\}$.

There are two paths from B to E .

Path $B \rightarrow D \rightarrow E$
is blocked by node D ,
satisfying condition (1).

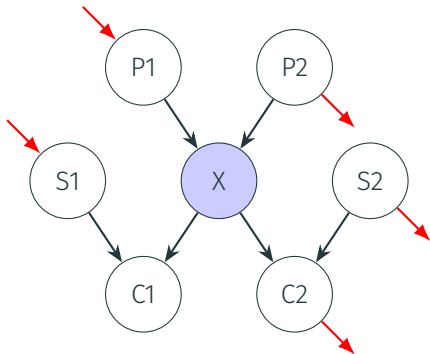
Path $B \rightarrow D \leftarrow A \rightarrow C \rightarrow E$
is not blocked by any node.

The statement is .



Markov Blanket

A **Markov Blanket** B_X of node X consists of **parents** of X , **children** of X and **"spouses"** (other parents of children of X , but not X) of X .



Every variable is conditionally independent of any other variable given it's **Markov Blanket**.

That's all folks!